



# Investigating behavior, attitude and intention towards waste segregation in tier II cities of India using theory of planned behavior

Maheshchandra Babu Jampala<sup>a</sup>, Tina Shivnani<sup>b,\*</sup>

<sup>a</sup> Department of Business Administration, Manipal University Jaipur, Rajasthan 303007, India

<sup>b</sup> Department of Commerce, Manipal University Jaipur, Rajasthan 303007, India

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## ABSTRACT

India's rapid urbanisation and waste rise make sustainable waste management challenging. Present study utilizes theory of planned behavior (TPB) to identify issues related to behavioral adaptations in waste segregation among the households' procedures in Jaipur and Indore cities of India. 448 respondents from each city were selected using convenience sampling. A self-designed questionnaire containing 30 items corresponding to attitude (AT), Perceived Behaviour Control (PBC), Subjective norms (SN) and Intention (IN) was used. The demographic profile of the respondents was also collected, and its analysis revealed nearly uniform distribution of the sample population. No significant effect of any of the demographic parameter was noted on either of the attributes. Respondent's city showed significant influence on all the 4 attributes due to drastic variations recorded in the responses for each determinant. SN had the biggest impact on IN, although all three hypotheses were validated and showed significantly beneficial impacts. The proposed model was a very good fit. Further, comparison of model for both cities revealed that only 2 attributes i.e. AT & SN in case of Jaipur and AT & PBC in case of Indore had significant positive effect on waste sorting intentions. The findings of this study provide a comprehensive comparison between two distinct cities, emphasizing how various behavioral factors contribute to changes in intention within diverse contexts. The key concerns that are highlighted based on the responses of residents from different cities could be used to prepare a roadmap for improving sensitivity towards waste segregation among Indian population.

## 1. Introduction

Urbanization and economic development in India have significantly increased waste production, adding strain to the country's already stressed waste management systems. With rapid population growth and expanding urban areas, managing solid waste has become an urgent issue. According to a study, India generates over 150 million tonnes of solid waste annually and this figure is expected to increase with increasing population and urbanization (Balasubramanian, 2018; Sharma and Jain, 2020). According to annual report of CPCB (Central Pollution Control Board) published in 2020–21, India produced 160,038.9 tonnes of solid waste per day (TPD) and ~152,749.5 TPD of solid waste was collected, out of which 50 % (79,956.3 TPD) was subjected to waste treatment and 18.4 % (29,427.2 TPD) was disposed off at landfill sites. However, 31.7 % (50,655.4 TPD) of the waste remained

unaccounted for emphasizing the urgent need for improved waste management to prevent environmental and health risks (CPCB, 2021). Further, as per year 2020 waste generation statistics, municipal solid waste generated in cities ranged from 0.2 kgs to 0.8 kgs per person per day and overall urban municipal waste is predicted to be around 0.5 kgs per person per day (Wilson, 2023). The increasing buildup of solid waste has created a pressing demand for sustainable and effective waste management solutions to tackle the environmental and public health issues that are linked to it. In fact, as per recent UN report of 2024, it is crucial for the world to transition from the era of solid waste and instead focus on transforming junk into a valuable resource.

Lack of proper infrastructure for solid waste collection, treatment and disposal in India poses serious risks to public health and the environment. Rapid urbanization and industrialization have worsened these challenges by contributing to pollution of air, water and soil, thus

\* Correspondence to: Department of Commerce, Manipal University Jaipur, Jaipur-Ajmer Express Highway, Dehmi Kalan, Near Toll Plaza, Jaipur, Rajasthan 303007, India.

E-mail addresses: [mahesh.jampala@jaipur.manipal.edu](mailto:mahesh.jampala@jaipur.manipal.edu), [jampalamahesh@gmail.com](mailto:jampalamahesh@gmail.com) (M.B. Jampala), [teena.shivnani@jaipur.manipal.edu](mailto:teena.shivnani@jaipur.manipal.edu), [teenashivnani13@gmail.com](mailto:teenashivnani13@gmail.com) (T. Shivnani).

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leading to significant health hazards (Arya and Kumar, 2020; Kalyanasundaram et al., 2021; Mahajan, 2023). Addressing these issues through effective waste management strategies such as waste segregation, is crucial for mitigating health risks and promoting environmental sustainability in Indian cities.

Waste segregation is the process of sorting waste at the source for effective recycling and disposal (Luckins et al., 2022; Roy et al., 2023). This approach has gained significant attention across the globe as it enables sustainable management of waste and minimizes the environmental impact. Globally, countries like Germany, Sweden, Japan, and South Korea have implemented effective waste segregation systems, leading to high recycling rates and minimal landfill use. In India, only a few cities like Mumbai, Bengaluru, Pune, and Chennai have implemented waste segregation practices effectively. However, studies show that due to improper segregation of municipal waste, only 60 % of the total plastic waste generated is recycled annually (Goh et al., 2022; Shanker et al., 2023; Susanth et al., 2021; Vincent, 2023). Waste segregation at the source remains a significant challenge in India, leading to inefficiencies in waste treatment and recycling processes. Several challenges including infrastructural, social, technological and regulatory have been associated with waste segregation in India (Table 1). Additionally, limited understanding of behavioral determinants such as attitudes, subjective norms, and perceived control in waste segregation, as well as a lack of comparative studies across different urban regions are some of the other research gaps specific to India. The influence of demographic factors remains underexplored and there is a need for context-specific interventions tailored to the unique challenges of regions like Rajasthan.

To address these research gaps, the Theory of Planned Behavior (TPB) can be employed to analyze the behavioral determinants influencing waste segregation. By examining attitudes, subjective norms, and perceived behavioral control, TPB provides a framework to understand and predict waste management behaviors, allowing for the design of targeted interventions tailored to the specific context of Rajasthan and other Indian regions. Therefore, this study introduces a novel application of the Theory of Planned Behavior (TPB) to the context of waste segregation in India, specifically focusing on two distinct urban environments -Jaipur and Indore. Unlike previous studies, this research provides a comparative analysis of behavioral determinants across these cities, revealing significant regional differences in how attitudes, perceived behavioral control, and subjective norms influence waste segregation intentions. The study also challenges the conventional understanding by showing that demographic factors have no significant effect on these behavioral attributes. By highlighting the varying impact of behavioral factors in different urban settings, this work offers new insights into the development of targeted, context-specific interventions for improving waste management practices in India.

## 2. Theoretical framework and hypothesis development

### 2.1. Theory of planned behavior (TPB)

The Theory of Planned Behavior (TPB) is a frequently used framework for examining the cognitive factors that influence health behaviors (Manning, 2009; Manstead and Parker, 1995; Sutton, 1997). The theory is an expansion of the Theory of Reasoned Action (TRA) and operates under the assumption that behavior is influenced by both the individual's purpose to engage in the behavior and their level of control over carrying out the behavior (Al Maskari, 2015). The TPB comprises the following essential elements:

- **Attitude towards behavior** pertains to an individual's favorable or unfavorable assessment of engaging in a specific behavior (Sharma et al., 2024).
- **Subjective norms** refer to an individual's beliefs on the approval or disapproval of key people in their life towards a certain behavior. It

**Table 1**

Different types of challenges associated with waste segregation in India.

| Challenges                 | Key Findings                                                                                                                                                                                                                                      | References                                                                                                                                                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Infrastructural Challenges | <b>Insufficient Waste Collection Systems</b><br>Merely 90 % of waste is gathered, only 20 % of the gathered waste is processed, while the other portion is disposed of in open landfills, resulting in environmental contamination.               | Prajapati et al. (2021), Sharma and Chandel (2021), Dutta and Goel (2021), Sahoo et al. (2022), Chaudhary et al. (2021), Pal and Bhatia (2022)                                                                                                           |
| Infrastructural Challenges | <b>Expensive Process</b><br>The expenses associated with waste management, such as the collection and transportation are more significant, which poses economic difficulties in implementing efficient waste segregation and disposal techniques. | Prajapati et al. (2021), Sharma and Chandel (2021), Khan et al. (2022), Dutta and Goel (2021), Thakur et al. (2021)                                                                                                                                      |
| Infrastructural Challenges | <b>Insufficient Infrastructure</b><br>Additional waste treatment facilities are required to accommodate the large quantity and varied kind of waste.                                                                                              | Khan et al. (2022), Dutta and Goel (2021), Malav et al. (2020), Priyadarshini and Abhilash (2020), Batra et al. (2022), Sengupta et al. (2023), Chaudhary et al. (2021), Ganguly and Chakraborty (2021)                                                  |
| Infrastructural Challenges | <b>Human resource limitations</b><br>Public waste collection services experience constraints in terms of available workforce, which impacts waste collection schedules and overall efficiency.                                                    | Kulkarni and Anantharama, 2020                                                                                                                                                                                                                           |
| Infrastructural Challenges | <b>Landfill Issues</b><br>The majority of landfills are either open dumping sites or partially regulated, resulting in significant methane emissions and uncontrolled fires.                                                                      | Chaudhary et al. (2021)                                                                                                                                                                                                                                  |
| Infrastructural Challenges | <b>Insufficient Organization and Financial Resources</b><br>Many cities encounter ongoing difficulties in delivering sustainable waste management services as a result of inadequate organization and financial resources.                        | Kalyanasundaram et al. (2021)                                                                                                                                                                                                                            |
| Social Challenges          | Public participation in municipal solid waste management is hindered by inadequate and ineffective integration between the public and commercial sectors.                                                                                         | Khan et al. (2022), Malav et al. (2020), Priyadarshini and Abhilash (2020), Chaudhary et al. (2021), Kalyanasundaram et al. (2021)                                                                                                                       |
| Social Challenges          | <b>Education and Awareness</b><br>It is necessary to provide instruction on the proper sorting and recycling of trash/solid waste to encourage improved practices in waste management.                                                            | Khan et al. (2022), Dutta and Goel (2021), Malav et al. (2020), Sahoo et al. (2022), Kulkarni and Anantharama (2020), Prajapati et al. (2021), Batra et al. (2022), Sengupta et al. (2023), Yadav et al. (2022), Soni et al. (2022), Dehal et al. (2022) |
| Social Challenges          | <b>Public Practices</b><br>Social practices and obstacles impede the adoption of specific technologies such as incineration.                                                                                                                      | Pal and Bhatia (2022), Sharma and Chandel (2021)                                                                                                                                                                                                         |
| Social Challenges          | <b>Health Hazards</b><br>Informal garbageman workers frequently face health risks as a result of inadequate waste handling practices and a lack of protective measures.                                                                           | Malav et al. (2020), Sengupta et al. (2023)                                                                                                                                                                                                              |

(continued on next page)

Table 1 (continued)

| Challenges               | Key Findings                                                                                                                                                                                       | References                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Social Challenges        | Social norms and stigma can cause individuals to experience embarrassment when segregating waste in the presence of their neighbors, as they fear being judged.                                    | Kalyanasundaram et al. (2021), Shanker et al. (2023)                                                                                                                            |
| Social Challenges        | The formal sector has limited involvement in waste management activities, resulting in a negative impact on the overall efficiency and efficacy of waste segregation.                              | Yadav et al. (2022), Ganguly and Chakraborty (2021)                                                                                                                             |
| Social Challenges        | <b>Public Attitude</b><br>It is essential to comprehend the public's stance on garbage waste generation and management to successfully carry out the necessary actions.                            | Thakur et al. (2021)                                                                                                                                                            |
| Social Challenges        | <b>Behaviour</b><br>Sustaining the modification of behavior in waste segregation procedures poses a considerable problem.                                                                          | Sahoo et al. (2022), Priyadarshini and Abhilash (2020), Batra et al. (2022), Prajapati et al. (2021), Ganguly and Chakraborty (2021), Thakur et al. (2021), Dehal et al. (2022) |
| Technological Challenges | The elevated humidity content of municipal solid waste (MSW) hinders the retrieval of recyclable materials and diminishes the effectiveness of waste-to-energy procedures.                         | Khan et al. (2022), Sharma and Chandel (2021), Soni et al. (2022), Ganguly and Chakraborty (2021)                                                                               |
| Technological Challenges | <b>Standards for the Recycling Industry</b><br>The recycling industry requires enhancements in professionalization, product standards, market expansion, and operating standards.                  | Khan et al. (2022), Dutta and Goel (2021), Kulkarni and Anantharama (2020), Chaudhary et al. (2021), Sengupta et al. (2023)                                                     |
| Technological Challenges | <b>Heterogeneity of Waste</b><br>The quality and quantity of recovered resources are inconsistent, affecting the economic viability of waste treatment systems.                                    | Priyadarshini and Abhilash (2020), Sharma and Chandel (2021)                                                                                                                    |
| Technological Challenges | Industrial symbiosis is essential for the efficient recycling of e-waste, but its implementation is currently constrained by limited collaboration between national and multinational enterprises. | Dutta and Goel (2021), Kulkarni and Anantharama (2020)                                                                                                                          |
| Technological Challenges | Lack of expertise in operating large-scale anaerobic digestion plants in India is a problem for implementing biological processes.                                                                 | Sahoo et al. (2022), Sharma and Chandel (2021)                                                                                                                                  |
| Technological Challenges | Operational problems encompass deficiencies in logistical planning, insufficient money, and a lack of knowledge in waste management technology.                                                    | Prajapati et al. (2021)                                                                                                                                                         |
| Regulatory Challenges    | <b>Uneven Execution</b><br>The level of development and application of MSW categorization schemes differs among different areas, hindering the creation of a unified national system standards.    | Khan et al. (2022), Dutta and Goel (2021), Malav et al. (2020), Dehal et al. (2022), Kulkarni and Anantharama (2020), Sengupta et al. (2023)                                    |
| Regulatory Challenges    | <b>Legal Framework</b><br>It is necessary to establish legal recognition for the involvement of the business                                                                                       | Khan et al. (2022), Yadav et al. (2022), Sahoo et al. (2022), Prajapati et al. (2021)                                                                                           |

Table 1 (continued)

| Challenges            | Key Findings                                                                                                                                                                                                                  | References                                                                                                                                                                                    |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | sector in managing municipal solid waste (MSW), and to include private stakeholders in larger collaborations standards.                                                                                                       |                                                                                                                                                                                               |
| Regulatory Challenges | The economic feasibility of MSW treatment systems is determined by several factors, such as governmental backing and financial incentives.                                                                                    | Ganguly and Chakraborty (2021); Sahoo et al. (2022); Sharma and Chandel (2021)                                                                                                                |
| Regulatory Challenges | Introducing innovative revenue sources such as user charges, tipping fees, and feed-in tariffs is necessary to enhance the financial sustainability of waste management systems.                                              | Sharma and Chandel (2021), Malav et al. (2020), Xie et al. (2024)                                                                                                                             |
| Regulatory Challenges | <b>Implementation of Regulations</b><br>A rigorous enforcement of regulations for the management of electronic waste is necessary to protect human health, the environment, and natural resources.                            | Priyadarshini and Abhilash (2020), Dutta and Goel (2021), Yadav et al. (2022), Dehal et al. (2022), Soni et al. (2022), Chaudhary et al. (2021), Sengupta et al. (2023), Thakur et al. (2021) |
| Regulatory Challenges | The current disaster waste management strategies are insufficient to address the distinct issues presented by pandemics, necessitating the use of optimisation and decision-making tools to enhance preparation and response. | Kulkarni and Anantharama (2020)                                                                                                                                                               |

also includes the individual's incentive to conform to those beliefs (Finlay et al., 1999).

- **Perceived behavioral control** refers to an individual's perception of their ability to manage their own behavior. This pertains to an individual's subjective assessment of how easy or challenging it is to carry out a particular behavior (Wallston et al., 1987).

The TPB model have been utilized in several studies to examine the elements that impact persons' intentions and actions regarding waste segregation (Akhter et al., 2024; Gokarn et al., 2023; Halder and Singh, 2018; Pratap et al., 2020; Sahoo et al., 2022). It has also been applied to investigate the factors that drive the workforce's engagement in construction waste management in India, with a particular focus on the importance of perceived behaviour control and knowledge (Tezeswi and MVN, 2022). Vijayan et al. (2023) applied TPB model to analyse the recycling behaviour of families in southern India regarding e-waste and reported that attitude, subjective norms, and perceived behavioral control have a substantial influence on both recycling intention and behavior.

Despite of wide application of TPB in waste segregation studies in India as well as abroad, there are certain limitations in this research area that need to be addressed, which includes the following:

- **Limited Context-Specific Studies:** Most studies that utilize TPB to analyze waste management carried out in nations with high levels of affluence. However, there is a scarcity of research that specifically examines developing countries like India. This leads to a lack of comprehension regarding the distinct socio-cultural and economic aspects that impact waste segregation in India (Raghu and Rodrigues, 2020).
- **Underemphasized Psychosocial Factors:** Empirical study on the psychosocial factors influencing solid waste segregation in India is insufficient. Research indicates that the outcomes of behavior have a

significant impact on waste segregation intents and actions, surpassing the importance of knowledge and attitude (Pratap et al., 2020).

- **Need for Comprehensive Models:** Many current researches fail to incorporate supplementary factors like motivation and perceived risk, which have the potential to increase the predictive accuracy by using comprehensive TPB model (Sujoed et al., 2023).
- **Evaluation Studies Deficit:** There is a lack of assessment studies that measure the effectiveness of interventions based on the TPB in real-world contexts. Most research is centered around the examination of models rather than the practical application and assessment of those models (Raghu and Rodrigues, 2020).
- **Youth and Educational Intentions:** Studies suggest that social factors play a crucial role in influencing the recycling intentions of young people in India. Nevertheless, further research is required to specifically investigate instructional strategies aimed at fostering waste segregation intentions among the youth (Halder and Singh, 2018).
- **Behavioral Change Components:** Research highlights the significance of motivation, opportunity, and ability in encouraging the adoption of sustainable waste segregation strategies. However, further research is required to determine the optimal methods for incorporating these elements into waste segregation strategies (Sahoo et al., 2022).

The present study utilizes the TPB (Theory of Planned Behavior) model to identify issues related to behavior adaptation in waste segregation among the households' in two cities of India.

### 3. Conceptual framework

Thus, based on these facts the objectives of the present study were to:

- compare the attitude, perceived behavior control and subjective norms towards household waste segregation among 2 cities (Jaipur and Indore) of India.
- determine the factors affecting the waste segregation intention among tire II cities in India.

Further, to achieve the above-mentioned objectives following hypotheses were formulated and a relationship model between the 4 factors of TPB (attitude, behavior, subjective norms and intention) as shown in Fig. 1 were also proposed:

**Hypothesis 1. (H1):** Attitude is positively related to the intention towards household waste segregation among 2 cities in India: Jaipur and Indore.

**Hypothesis 2. (H2):** Perceived behavior control is positively related to the intention towards household waste segregation among 2 cities in India: Jaipur and Indore.

**Hypothesis 3. (H3):** Subjective Norms are positively related to the intention towards household waste segregation among 2 cities in India: Jaipur and Indore.

The model was used because of the following reasons: first, research has demonstrated that attitude has a crucial role in predicting the desire to segregate waste among diverse populations, such as students and

people in different geographical areas (Setiawan et al., 2020). Some studies, it was discovered that perceived behavioural control has a more significant impact on behavioural intention compared to characteristics such as attitude and subjective norms (Wang and Mangmeechai, 2021; Yu et al., 2018). Subjective norms have been identified as a strong predictor of waste segregation intention in certain research, however their influence may vary based on the specific situation and demographic (Costa et al., 2022; Setiawan et al., 2020).

## 4. Data and methodology

### 4.1. Questionnaire design

Data collection was done using a questionnaire which was prepared using a seven-step process illustrated in Fig. 2.

The questionnaire design was divided into two separate sections: (i) the first part of the questionnaire pertains to the demographic information of the respondents, encompassing factors such as gender, age, marital status, occupation, and educational background of the respondents, and (ii) the second section pertains to the behaviour of waste separation at its source and discusses the additional factors that influence TPB determinants including attitude, subjective norm and perceived behavioural control. Initially, a total of 50 questions of TPB model were prepared, out of which only 30 questions were finally selected for final data collection after comments from the experts. The final questionnaire contained 9, 7, 7 and 7 questions related to the four attributes of TPB i.e. attitude, perceived behavior, subjective norms and intention, respectively (Supplementary Material). The reliability of the questionnaire was assessed for each attribute separately using Alpha Cronbach method as reported by Ghani et al. (2013), and the reliability coefficient ( $\alpha$ ) for attitude, perceived behavior, subjective norms, and intention was 0.846, 0.769, 0.811 and 0.733, respectively. Previous reports have indicated that, reliability coefficient above 0.7 is an indicator of good reliability (Fan et al., 2019; Vijayan et al., 2023). Razali et al. (2020) reported that high reliability coefficient represented that all the constructs represented their respective latent variables i.e. attributes of TPB very well.

### 4.2. Selection of study area

In this study, two cities were chosen based on the rankings of different cities in Swachh Survekshan (2023). The Swachh Survekshan is a yearly assessment that evaluates the level of cleanliness, hygiene, and sanitation in urban regions of India. The survey is carried out by the Ministry of Housing and Urban Affairs (MoHUA) with the objective of motivating cities to enhance their sanitation and cleanliness standards. Evaluation of cities during Swachh Survekshan (2023) was done on the basis of 3 primary components: (i) service level progress, (ii) certificate, and (iii) citizen voice which constituted 51 %, 26 % and 23 % of the total score, respectively. As per the Swachh Survekshan (2023) rankings, Indore city had highest score among all cities the state Madhya Pradesh (MP) and it ranked 1st nationwide (Table 2). Further, nationwide ranking of Jaipur city was 171 (Heritage region) and 173 (Greater region), however it had highest score among all cities of Rajasthan State (Table 3). The Survekshan data also showed that % of waste segregated at the source was ~7.25 % (Heritage region) and 23.25 % (Greater region) in Jaipur city in comparison to 98.13 % in the top ranked city i.e. Indore (Tables 2 and 3). Therefore, based on these statistics, Jaipur city was chosen as one of the study areas as it is a popular destination for tourists and Indore city was chosen (used as standard) for comparison.

### 4.3. Sampling method and data collection

In the present study, selection of sample population from both chosen cities was done using convenience sampling method as recommended by Jager et al. (2017). Convenience sampling is a

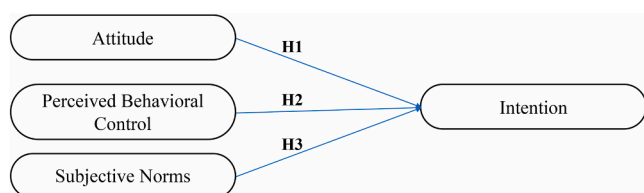


Fig. 1. Research model.

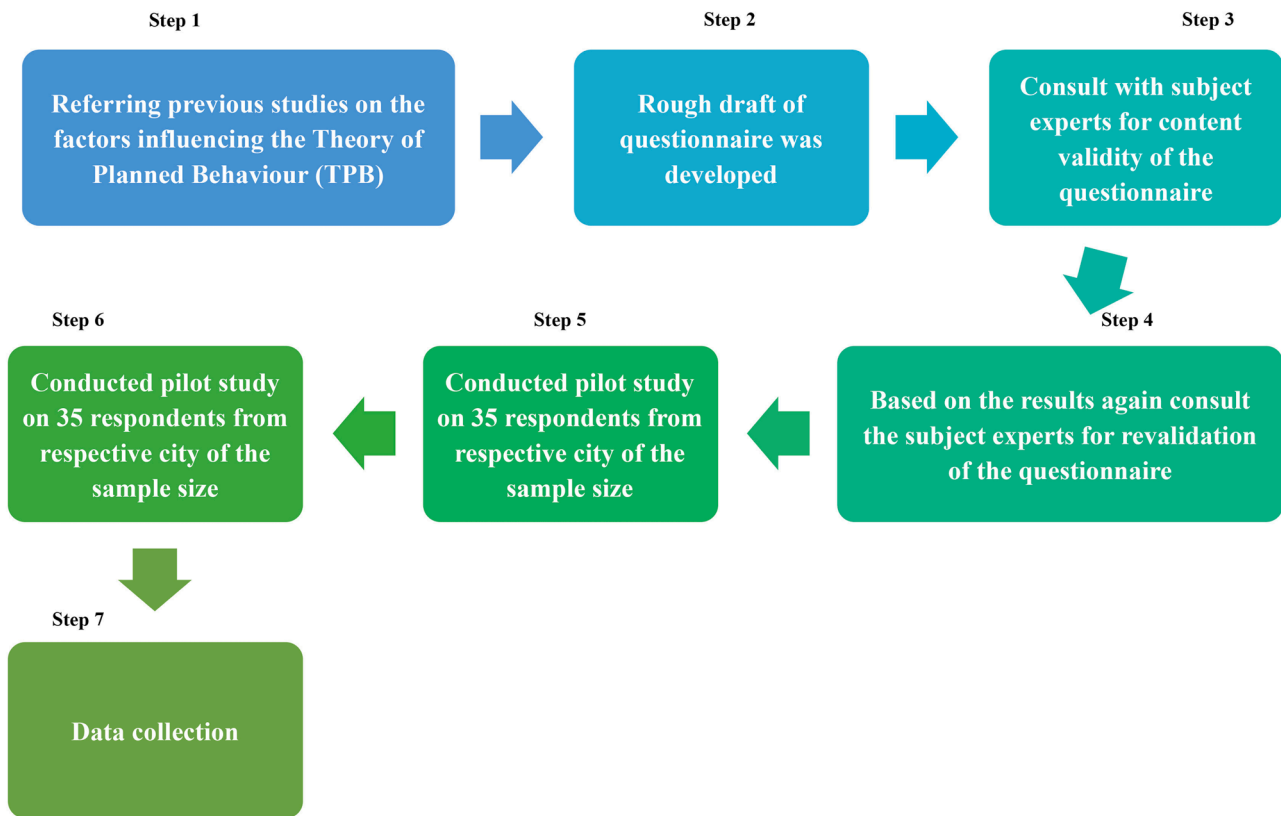


Fig. 2. Stepwise process followed for preparation of questionnaire for data collection.

Table 2

Summary of top 10 Urban Local Bodies (ULB) of Madhya Pradesh state as per Swachhta Survekshan 2023.

| S. No | ULB Name | Population | % D2D | % Source Segregation | % Processing | SR | AR | NR |
|-------|----------|------------|-------|----------------------|--------------|----|----|----|
| 1     | Indore   | > 10 L     | 99.6  | 98.1                 | 100          | 1  | 1  | 1  |
| 2     | Bhopal   | > 10 L     | 99.6  | 95                   | 100          | 2  | 5  | 5  |
| 3     | Budni    | 15–25k     | 99.5  | 45                   | 100          | 1  | 33 | 8  |
| 4     | Khurai   | 50k - 1 L  | 99.5  | 100                  | 88.7         | 2  | 51 | 15 |
| 5     | Gwalior  | > 10 L     | 97.7  | 98                   | 70.6         | 3  | 19 | 16 |
| 6     | Dhar     | 50k - 1 L  | 99.5  | 100                  | 99.9         | 3  | 53 | 16 |
| 7     | Sausar   | 25–50k     | 96.5  | 98                   | 50           | 4  | 55 | 17 |
| 8     | Mungaoli | 25–50k     | 98    | 100                  | 86.4         | 5  | 56 | 18 |
| 9     | Ujjain   | 3 L - 10 L | 100   | 100                  | 86.1         | 4  | 22 | 19 |
| 10    | Rau      | 25–50k     | 99    | 99                   | 50           | 6  | 57 | 19 |

SR – State Rank; AR – Absolute Rank; NR – National Rank

Source: [Swachh Survekshan \(2023\)](#)

Table 3

Summary of top 10 ULBs of Rajasthan state as per Swachhta Survekshan 2023.

| S. No | ULB Name        | Population | % D2D | % Source Segregation | % Processing | SR | AR   | NR  |
|-------|-----------------|------------|-------|----------------------|--------------|----|------|-----|
| 1     | Jaipur Heritage | > 10 L     | 84.6  | 7.2                  | 0            | 1  | 988  | 171 |
| 2     | Jaipur Greater  | > 10 L     | 93.1  | 23.2                 | 15.8         | 2  | 991  | 173 |
| 3     | Udaipur         | 3 L - 10 L | 80.9  | 42.7                 | 74.2         | 3  | 1241 | 206 |
| 4     | Jodhpur South   | 3 L - 10 L | 92.9  | 9.6                  | 75.5         | 4  | 1329 | 210 |
| 5     | Ajmer           | 3 L - 10 L | 92.5  | 13                   | 0            | 5  | 1358 | 214 |
| 6     | Kota North      | 3 L - 10 L | 96.7  | 71.9                 | 11.2         | 6  | 1660 | 244 |
| 7     | Kota South      | 3 L - 10 L | 88.9  | 25.1                 | 0            | 7  | 1789 | 256 |
| 8     | Sikar           | 1 L - 3 L  | 58.5  | 0.2                  | 24.5         | 8  | 2138 | 281 |
| 9     | Baran           | 1 L - 3 L  | 98.2  | 43                   | 30.4         | 9  | 2149 | 283 |
| 10    | Bharatpur       | 1 L - 3 L  | 95.5  | 18 %                 | 0            | 10 | 2165 | 285 |

SR – State Rank; AR – Absolute Rank; NR – National Rank

Source: [Swachh Survekshan \(2023\)](#)

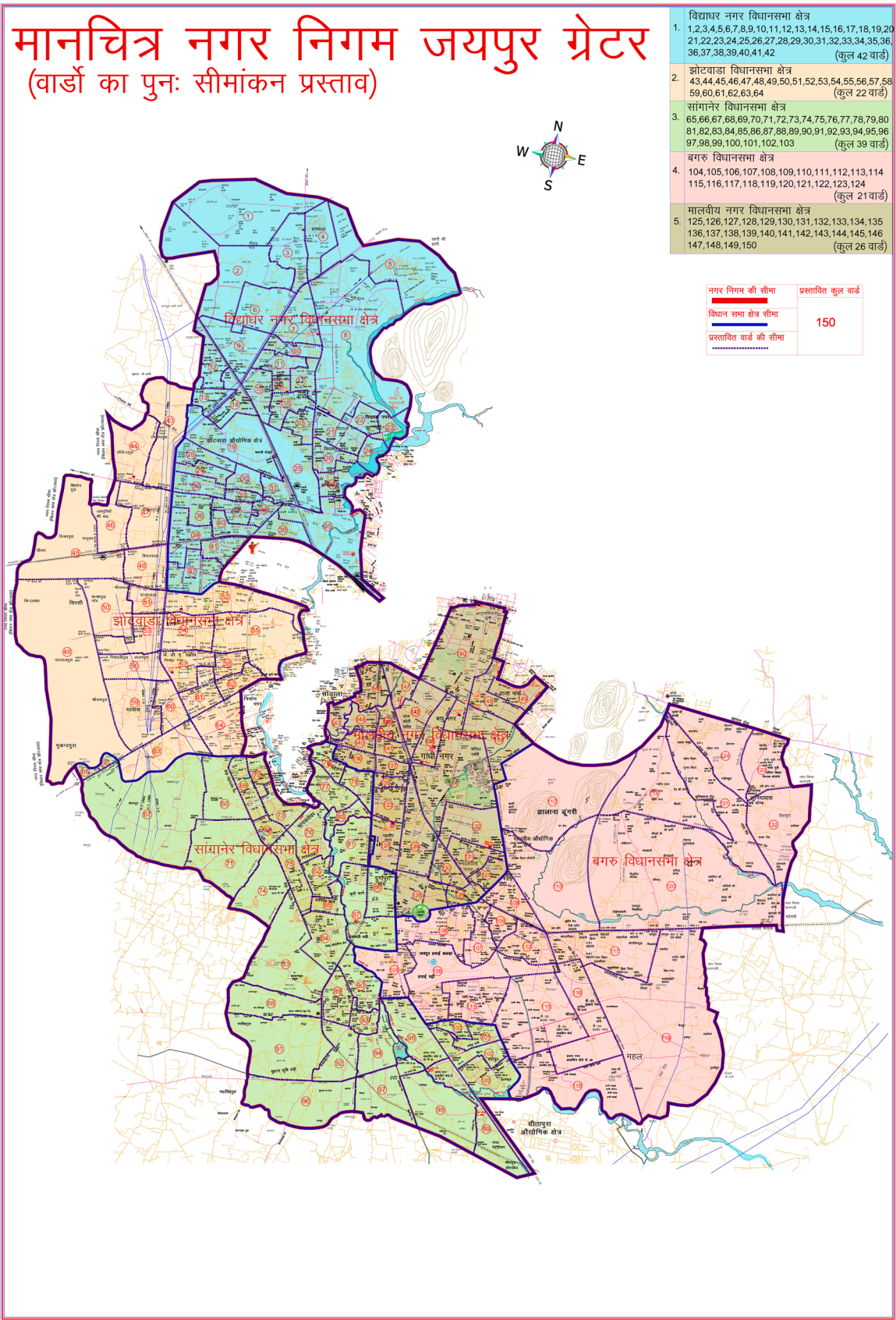


Fig. 3. Ward-wise map of Jaipur city as available on Jaipur Development Authority Website.

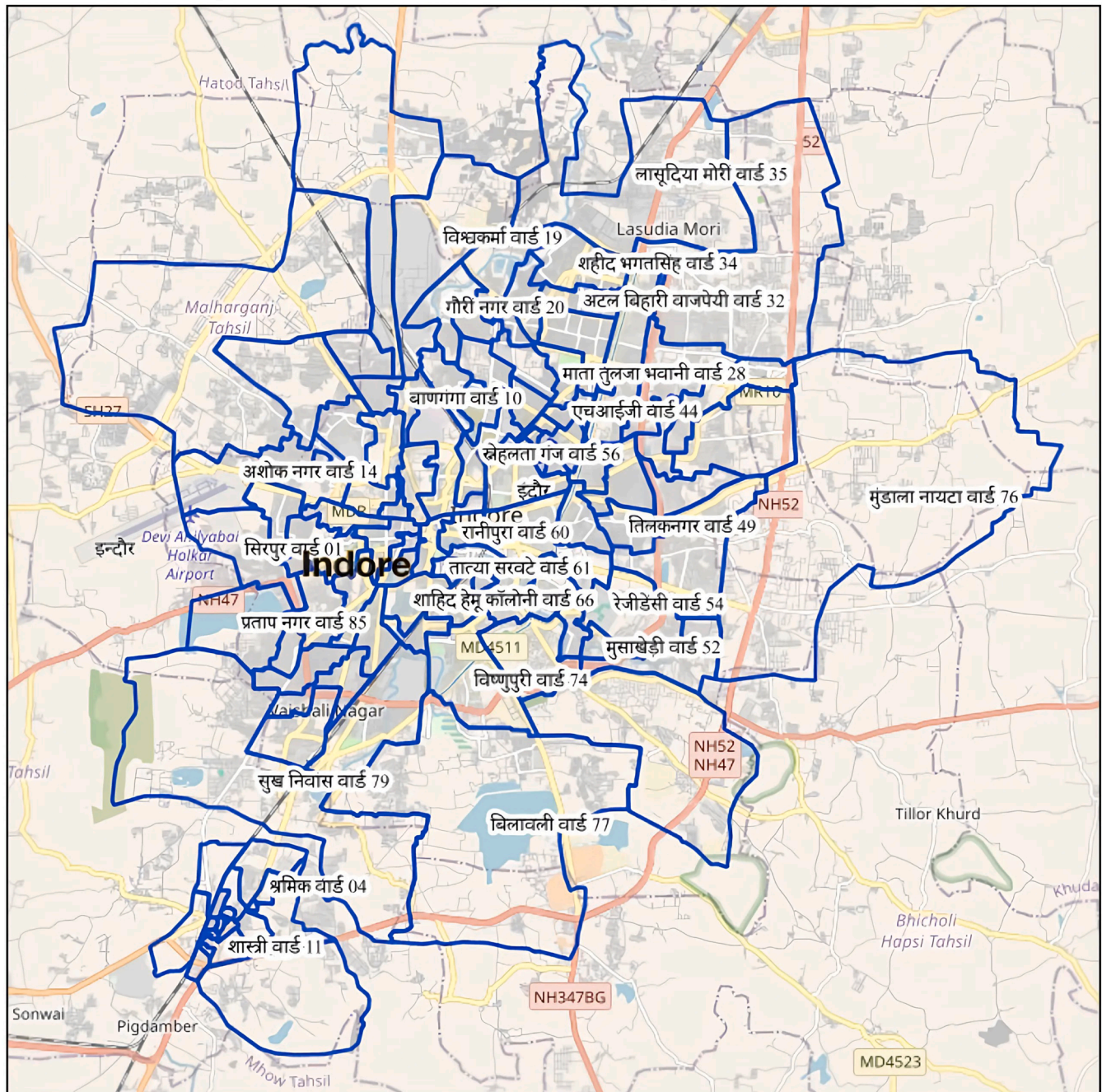


Fig. 4. Ward-wise map of indore city as available on MP Geo Portal Website.

non-probability sampling method that involves selecting respondents who are easily accessible, available, and willing to participate in the research (Dornyei, 2007). From the total 150 wards of Jaipur city and 85 wards of Indore city (Figs. 3 and 4), 10 wards having maximum number of households were identified in both cities and were then located for selection of respondents (Table 4). The number of households in 10 selected wards of Jaipur city ranged from ~11500 to ~18500, while in case of Indore city the number ranged from ~8000 to ~11500 (Table 4). Equal number of respondents (44 or 45) were selected from each ward such that total 448 (n) respondents were selected from both cities for data collection so as to eliminate biasness due to non-homogenous population distribution in terms of number of participants from each zone. Ward wise number of respondents selected in both cities is given in Table 4.

Further, the data was collected from selected respondents (sample population) through scheduling method using the structured survey. The scheduling method involves use of a structured questionnaire which ensured precise data collection during the research. As described earlier, the first part of the questionnaire collected demographic information about the respondents including gender, age, marital status, occupation and education level. The second part consisted of 30 questions representing different attributes to be studied. The data was collected simultaneously from both the cities for a period of 6 months (December 2023 – May 2024). The data thus collected was then subjected to several statistical tools for analysis.

**Table 4**

Ward-wise number of households (as per 2011 census) and selected respondents from Jaipur and Indore cities.

| Ward No                               | Total No. of House Holds | Total No. of Respondents Selected |
|---------------------------------------|--------------------------|-----------------------------------|
| <b>Jaipur city</b>                    |                          |                                   |
| Jaipur (M Corp.) (Part) WARD NO.-0011 | 18508                    | 45                                |
| Jaipur (M Corp.) (Part) WARD NO.-0031 | 17458                    | 45                                |
| Jaipur (M Corp.) (Part) WARD NO.-0034 | 15328                    | 45                                |
| Jaipur (M Corp.) (Part) WARD NO.-0001 | 13937                    | 45                                |
| Jaipur (M Corp.) (Part) WARD NO.-0029 | 13685                    | 45                                |
| Jaipur (M Corp.) (Part) WARD NO.-0032 | 13167                    | 45                                |
| Jaipur (M Corp.) (Part) WARD NO.-0002 | 12700                    | 45                                |
| Jaipur (M Corp.) (Part) WARD NO.-0023 | 12429                    | 45                                |
| Jaipur (M Corp.) (Part) WARD NO.-0035 | 11895                    | 44                                |
| Jaipur (M Corp.) (Part) WARD NO.-0007 | 11614                    | 44                                |
| <b>Indore city</b>                    |                          |                                   |
| Indore (M Corp.) WARD NO.-0042        | 11561                    | 45                                |
| Indore (M Corp.) WARD NO.-0050        | 11341                    | 45                                |
| Indore (M Corp.) WARD NO.-0022        | 10679                    | 45                                |
| Indore (M Corp.) WARD NO.-0059        | 9247                     | 45                                |
| Indore (M Corp.) WARD NO.-0019        | 9087                     | 45                                |
| Indore (M Corp.) WARD NO.-0002        | 9029                     | 45                                |
| Indore (M Corp.) WARD NO.-0034        | 8511                     | 45                                |
| Indore (M Corp.) WARD NO.-0037        | 8387                     | 45                                |
| Indore (M Corp.) WARD NO.-0040        | 8025                     | 44                                |
| Indore (M Corp.) WARD NO.-0058        | 8017                     | 44                                |

Source: <https://censusindia.gov.in/census.website/data/census-tables>

#### 4.4. Data analysis using statistical tools

The data analysis was performed using two software: IBM's Statistical Package for Social Sciences (SPSS, v23.0) and Smart PLS 4 based on the approach adopted in previously published similar studies by Razali et al. (2020) and Sharma et al. (2024). Reliability of the questionnaire items was determined on the basis of their Cronbach's alpha coefficients using SPSS. Descriptive statistics was used to describe socio-demographic characteristics of the sample population (respondents) using SPSS. Frequency distribution tables and measure of central tendency (mode) were used to describe the responses obtained for each question of the construct/instrument representing different attributes (attitude, behavior, subjective norms and intention). Pearson's correlation using SPSS was performed to determine inter-relationship between different attributes (among different items of each attribute and overall attributes) and chi-square test was performed to determine effect of different socio-demographic parameters on attitude, behavior, intention and subjective norms for validation of the proposed hypothesis. Further, partial least squares structural equation modelling (PLS-SEM) using Smart PLS was performed to test the proposed model. The PLS-SEM algorithm was run along with bootstrapping on default settings and confidence interval method set to bias corrected & accelerated (BCa) as performed by Razali et al. (2020).

## 5. Results

### 5.1. Socio-demographic characteristics of sample population

Analysis of socio demographic data collected from respondents was done to identify the characteristics of the selected sample population in terms of age, gender, marital status, education level and occupation as summarized in Table 5. Majority (52 %) of the respondents were female and 46 % were male, amongst which most of them aged between 20 yrs to 30 yrs (23.8 %) and 40 yrs to 50 yrs (20.4 %), where in 19.6 %, 18.6 % and 17.5 % of them were having age between 30 yrs - 40 yrs, 60 yrs - 70 yrs and 50 yrs - 60 yrs, respectively. Only 228 respondents (25.4 %) of 896 were unmarried and majority (74.6 %) were married (Table 5). As far as the educational and professional background of the respondents is concerned, 27.6 % (247), 26 % (233), 24.3 % (218), 22.1 % (198) had qualified higher secondary, post-graduation, secondary and graduation, respectively and were either not working/working from home (14.6 %) or were having a private job (30.1 %), government job (28 %) or family business (27.2 %).

Cross-tabulation was also performed to check and compare the characteristics of respondents from Jaipur and Indore cities (Fig. 5). Proportion of male and female respondents in Jaipur city was nearly equal (50.7 % - female, 49.3 % - male), wherein case of Indore city proportion of female respondents (53.3 %) was higher than that of male respondents (46.6 %). The proportion of respondents between 30 years to 60 years age were nearly same in both Jaipur (58.7 %) and Indore cities (58.9 %), however the proportion of respondents having age range between 20 yrs - 30 yrs and 60 yrs - 70 yrs was 19.2 % and 22.1 %, respectively in Jaipur city and 23.4 % and 17.6 %, respectively in Indore city (Fig. 5). Marital status of the respondents also varied in Jaipur city and Indore city such that 23.2 % and 27.7 %, respectively were unmarried and 76.8 % and 72.3 %, respectively were married. Further, maximum proportion of respondents from Jaipur city were post-graduate (28.3 %) and from Indore city were higher secondary qualified (28.8 %) indicating difference in the educational background of the sample population of both cities thus selected. Further, in case of occupation characteristics maximum proportion of the respondents from Jaipur city were working in a private company (30.8 %), and from Indore city were working either in a private company (29.5 %) or had a government job (31.0 %).

Further, chi-square test revealed that the socio-demographic characteristics of respondents were independent of the city/subject area i.e. no significant association of any of the socio-demographic parameters with the city was detected (Table 6). Significant association of age of

**Table 5**

Scio-demographic characteristics of the respondents/sample population (n = 896).

| Characteristics | Category        | Frequency | % (total sample population) |
|-----------------|-----------------|-----------|-----------------------------|
| Age (in yrs)    | 20 - 30         | 213       | 23.8                        |
|                 | 30 - 40         | 176       | 19.6                        |
|                 | 40 - 50         | 183       | 20.4                        |
|                 | 50 - 60         | 157       | 17.5                        |
|                 | 60 - 70         | 167       | 18.6                        |
| Gender          | Female          | 466       | 52.0                        |
|                 | Male            | 430       | 48.0                        |
| Marital Status  | Unmarried       | 228       | 25.4                        |
|                 | Married         | 668       | 74.6                        |
| Occupation      | Private Job     | 270       | 30.1                        |
|                 | Family Business | 244       | 27.2                        |
|                 | Home Working    | 131       | 14.6                        |
|                 | Government Job  | 251       | 28.0                        |
| Education Level | Secondary       | 218       | 24.3                        |
|                 | Higher          | 247       | 27.6                        |
|                 | Secondary       |           |                             |
|                 | Graduation      | 198       | 22.1                        |
|                 | Post-Graduation | 233       | 26.0                        |

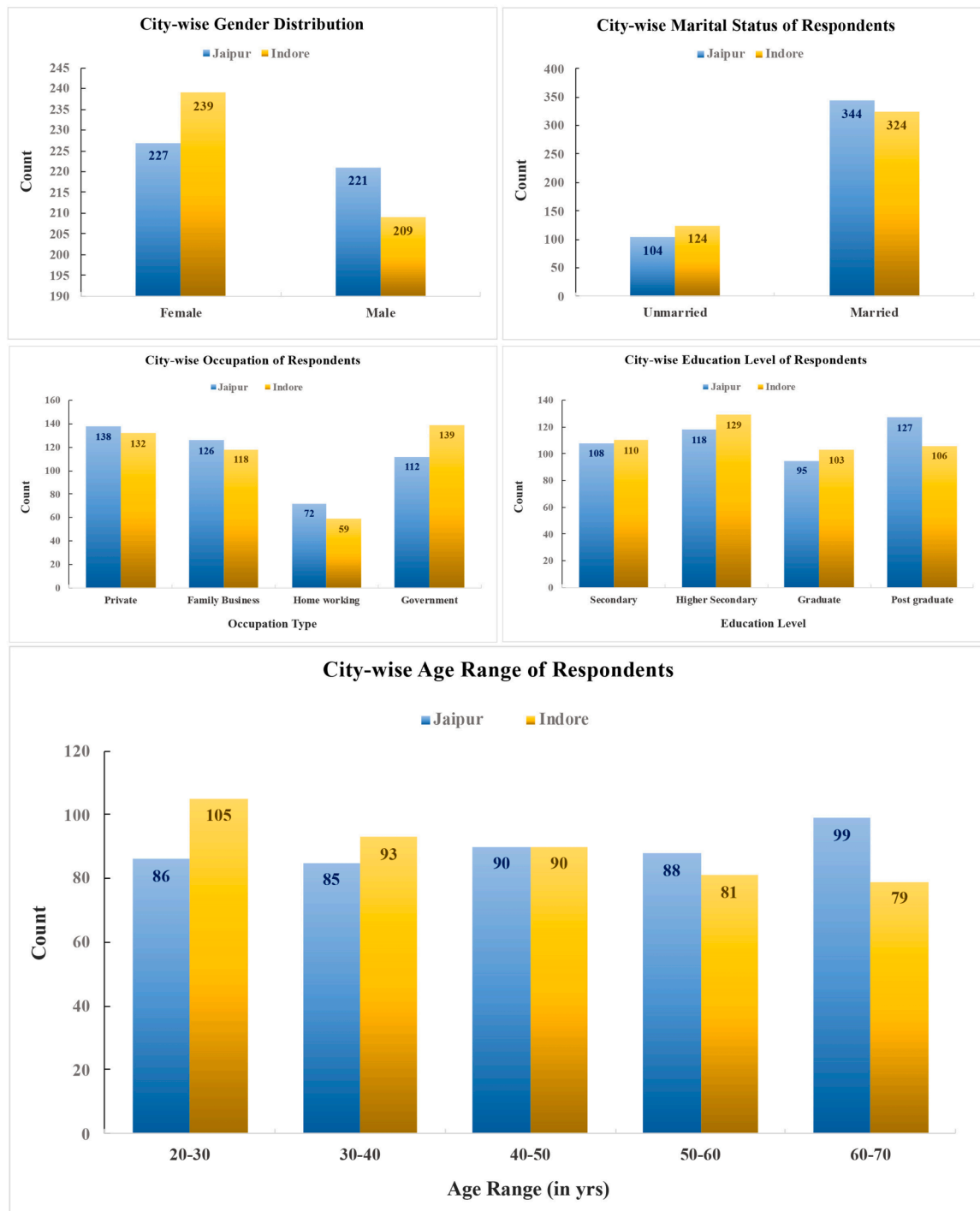


Fig. 5. City-wise socio-demographic characteristics of respondents.

respondents was noted with their occupation and marital status and that of occupation was noted with education level and gender, while no significant association was found between/among other parameters (Table 6).

## 5.2. Attributes of theory of planned behavior

### 5.2.1. Measurement model testing

Reliability and validity of the questionnaire items was tested in terms of Cronbach's alpha (CA), Composite reliability (CR), Outer loadings (OL) and Average variance extracted (AVE) using SPSS. Values of CA and CR of all the 4 latent variables (attitude, behavior, subjective norms

& intention) ranged between 0.710 and 0.949 and 0.740–0.958, respectively (Table 7). CA values above 0.7 indicated good reliability and CR values above 0.7 indicated good internal consistency of all the 4 attributes of the questionnaire. Further, values of AVE and OL for each latent variable was determined to measure the degree of variance in their respective indicators. AVE values were higher than 0.5 for all the attributes and it ranged between 0.632 and 0.768, (Table 7), which indicated that each attribute accounted for at least 50 % variance of its determinants and thereby affirming the reliability of the measurement construct. OL for each indicator of the latent variable was calculated and those having value less than recommended 0.6 were excluded to ensure adequate quality of structural model (Table 7). OL of all the

**Table 6**

Chi-square statistic of socio-demographic parameters.

| Variables tested                 | Pearson Chi-Square Value | P-value |
|----------------------------------|--------------------------|---------|
| City & Age                       | 5.474                    | 0.242   |
| City & Gender                    | 0.426                    | 0.514   |
| City & Marital Status            | 1.690                    | 0.194   |
| City & Education Level           | 3.048                    | 0.384   |
| City & Occupation                | 4.512                    | 0.211   |
| Age & Gender                     | 1.355                    | 0.852   |
| Age & Marital Status             | 823.672                  | 0.000*  |
| Age & Education Level            | 9.671                    | 0.645   |
| Age & Occupation                 | 22.554                   | 0.032*  |
| Gender & Marital Status          | 0.692                    | 0.405   |
| Gender & Education Level         | 0.820                    | 0.845   |
| Gender & Occupation              | 55.215                   | 0.000*  |
| Marital Status & Education Level | 2.876                    | 0.411   |
| Marital Status & Occupation      | 1.027                    | 0.795   |
| Occupation & Education Level     | 18.230                   | 0.033*  |

\* Significant values at 5 % significance level.

**Table 7**

Reliability and validity of the measurement model.

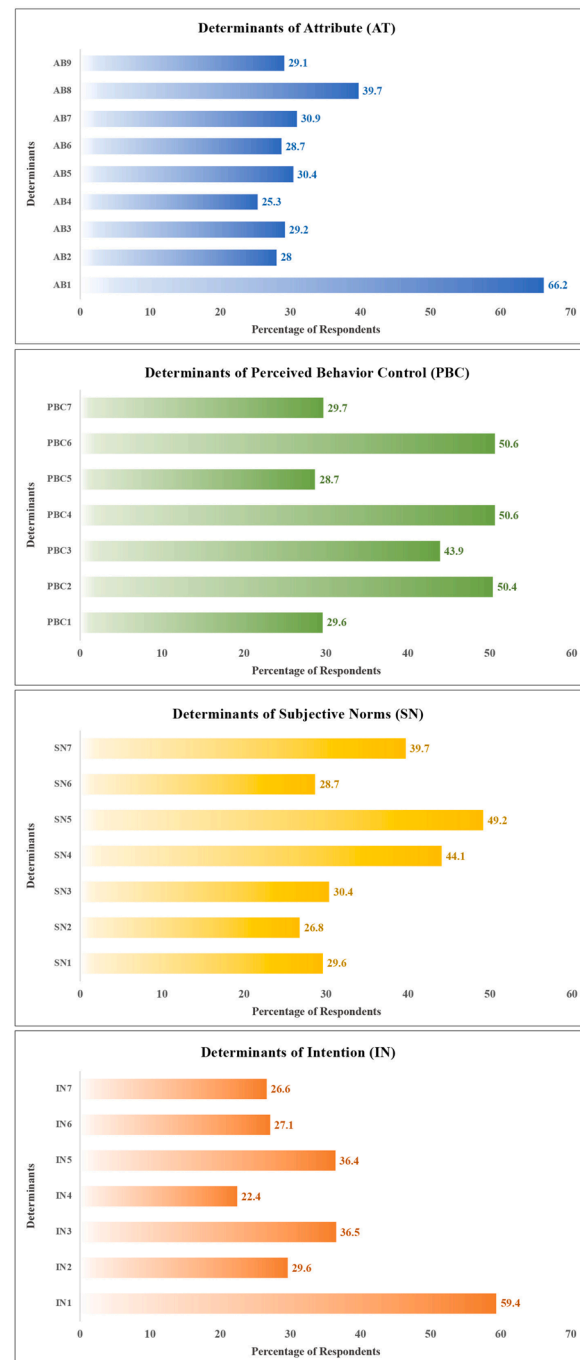
| Attributes<br>(Latent<br>variable)     | Determinants | OL    | CA    | CR -<br>$\rho_c$ | CR -<br>$\rho_a$ | AVE   |
|----------------------------------------|--------------|-------|-------|------------------|------------------|-------|
| Attitude (AT)                          | AT2          | 0.877 | 0.949 | 0.958            | 0.953            | 0.768 |
|                                        | AT3          | 0.906 |       |                  |                  |       |
|                                        | AT4          | 0.804 |       |                  |                  |       |
|                                        | AT5          | 0.868 |       |                  |                  |       |
|                                        | AT6          | 0.899 |       |                  |                  |       |
|                                        | AT7          | 0.918 |       |                  |                  |       |
|                                        | AT9          | 0.855 |       |                  |                  |       |
|                                        | PBC1         | 0.861 |       |                  |                  |       |
|                                        | PBC2         | 0.627 |       |                  |                  |       |
| Perceived<br>Behavior<br>Control (PBC) | PBC5         | 0.894 | 0.824 | 0.884            | 0.862            | 0.661 |
|                                        | PBC7         | 0.842 |       |                  |                  |       |
|                                        | SN1          | 0.831 |       |                  |                  |       |
| Subjective<br>Norms (SN)               | SN3          | 0.934 | 0.891 | 0.925            | 0.911            | 0.755 |
|                                        | SN5          | 0.768 |       |                  |                  |       |
|                                        | SN7          | 0.932 |       |                  |                  |       |
|                                        | IN2          | 0.867 |       |                  |                  |       |
| Intention (IN)                         | IN4          | 0.731 | 0.710 | 0.837            | 0.740            | 0.632 |
|                                        | IN8          | 0.781 |       |                  |                  |       |

determinants of attribute, behavior, subjective norms and intention ranged between 0.855 and 0.918, 0.627–0.894, 0.768–0.934 and 0.731–0.867, respectively. These values indicated that all the determinants used for their respective attributes were acceptable and reliable for the model.

### 5.2.2. Descriptive of TPB model attributes

Frequency distribution of responses recorded for each determinant/indicator of the latent variables was measured in order to identify the overall attitude, behavior, subjective norms and intention of the respondents towards waste segregation. The most common responses for each determinant was identified using standard measure of central tendency (mode) recommended for categorical variables (Fig. 6).

For latent variable 'Attitude (AT)', Most (66.2 %) of the respondents agreed that role of youth in waste segregation was "important" (AB1), wherein mixed opinion was recorded for the satisfaction level with the current waste collection & segregation services provided by the respondent's local authority, such that 45.9 % were 'satisfied' & 47.2 % were 'dissatisfied' (AB2). 29.2 % believed that there was 'no impact' of waste segregation on landfill usage in their community (AB3) and somewhat mixed response was noted for AB4, in which 25.3 %, 23.5 % and 21.2 % found the current educational materials available to public on waste segregation 'not very effective', 'somewhat effective' and 'neither effective nor ineffective', respectively. Similar to AB4, for AB5 also respondents expressed mixed opinion however most (30.4 %) of



**Fig. 6.** Most common responses along with their frequency % for determinants of all 4 attributes.

them did not found the current educational programs on waste segregation in influencing youth behavior very effective. 45.2 % of respondents agreed that they understand the environmental impact of improper waste segregation well, while 47.9 % either understand it poorly or not at all (AB6). Online resources were rendered as most effective (30.9 %) way to teach children about waste segregation (AB7). Majority (29.9 %) of the respondents disagreed that waste segregation can significantly reduce the cost of waste management (AB8) and 45.2 % respondents agreed on positive impact of waste segregation on their community's health (AB9).

For latent variable 'Behavior (PBC)', respondents agreed that the main reasons for not practicing waste segregation in household were either lack of motivation (29.6 %), time constraints (25.7 %) or

inadequate facilities (21.7 %). Majority (50.4 %) agreed that lack of government support, infrastructure, awareness and community participation were responsible for ineffective waste segregation in their local community (PBC2). 43.9 % respondents suggested that 'inadequate training for staff' is the biggest challenge in implementing successful waste segregation programs in schools and universities (PBC3). Half (50.6 %) of the sample population stated that "lack of support from other household members" is the biggest challenge that they face while trying to segregate waste at home (PBC4), and "lack of space" as the biggest challenge in implementing effective waste segregation in educational institutes (PBC6). Yet, more than half (52.1 %) respondents rated the segregation facilities in their area from fair to excellent (PBC 5). Regarding type of waste-segregation system, most (29.7 %) had no preference while 26.2 % and 23.1 % rendered multi-stream recycling and dual-stream recycling, as most user-friendly approaches (PBC7).

For latent variable 'Subjective norms (SN)', 29.6 % of the respondents said that educational institutes in their area never promote waste segregation practices (SN1) and 30.4 % refused of receiving any information about waste segregation from any source (SN2). Though, some of them did mention about receiving the information on waste segregation either monthly (26.8 %) or weekly (23.1 %). Similarly, though majority (30.4 %) of them were not sure about the differences in waste management practices of urban and suburban areas, while some presented contradicting opinion such that according to 25.6 % respondents there was no difference and as per 21.5 % and 18.2 % there was either slight or significant difference, respectively (SN3). Regarding role of local government in waste segregation, 44.1 % suggested their role in education and awareness campaigns wherein 33.7 % suggested their role in providing resources & facilities (SN4). Nearly half (49.2 %) of the respondents chose social media as their main/primary source of information regarding waste segregation practices (SN5). Based on responses of sample population from both the cities, it was also deduced that the frequency at which local authorities check/enforce waste segregation laws in their jurisdiction area varied from never *i.e.* zero (28.7 %) to occasionally (25.2 %) & very frequently (20 %) (SN6). Maximum respondents agreed that NGO's played a 'slight active' (39.7 %) role in promoting waste segregation practices in their locality (SN7).

For latent variable 'Intention (IN)', more than half proportion (55.1 %) of the respondents believed that both financial incentives (30 %) and community-based initiatives (29.4 %) could increase participation of households in waste segregation (IN1). 40 % agreed that presence of community leaders and volunteers to promote waste segregation have influential impact on waste segregation behavior, while 30 % disagreed to it (IN2). Despite of the fact that 41.2 % agreed that legal requirements is the main motivation/driving force behind their participation in waste segregation (IN3), yet according to 36.5 % respondents (maximum proportion), local competitions could increase participation in waste segregation (IN4). In line with the response for IN3, 36.4 % respondents thought that implementation of more stringent laws could improve waste segregation behavior in their community (IN5). Major proportion (74.9 %) of the respondents felt that incentives like monetary rewards (27.1 %), recognition/awards (24.3 %) and lower waste disposal fee (23.4 %) could promote waste segregation more diligently (IN6). Regarding disposal of electronic waste, only 20.9 % opted special collection bins whereas 22.8 % and 26.6 % either returned such items to retailers or disposed of in/along with regular solid wastesolid waste, respectively (IN7).

### 5.2.3. Effect of socio-demographic properties on TPB model attributes

Chi-square test was performed to check the influence of socio-demographic properties of the respondents on their response for each of the determinant and the same has been summarized below:

#### 5.2.3.1. Effect of age. All determinants of attitude except AT4, did not

exhibited any significant association with the age of the respondents, thereby indicating that the attitude of respondents towards waste segregation was irrespective of their age group (for majority of the determinants). Contrastingly, neither of the PBC, SN and IN determinants showed any significant association with the age of the respondents, indicating that the responses for all the determinants were not affected by the respondent's age group.

**5.2.3.2. Effect of gender.** No significant association of gender with any of the determinants of attitude, perceived behavior control, subjective norms & intention was detected, thereby indicating that respondent's attributes towards waste segregation did not differ among male and female respondents.

**5.2.3.3. Effect of marital status.** All determinants of attitude except AT4, did not exhibited any significant association with the marital status of the respondents, thereby indicating that the attitude of both married and unmarried respondents towards waste segregation did not significantly differ on majority of its determinants. On the contrary, none of the determinants of PBC, SN and IN exhibited significant association with marital status of the respondents, thus ruling out the chances of response biasness due to marital status.

**5.2.3.4. Effect of education level.** No significant association of education with any of the 9 determinants of the latent variable 'attitude' was detected, thereby indicating that education had no significant effect on the respondent's attitude towards waste segregation. Similarly, no significant association of any of the 7 determinants of perceived behavior control, subjective norms and intention with the educational qualifications of the respondents was detected.

**5.2.3.5. Effect of occupation.** All determinants of attitude except AT2, did not exhibited any significant association with the occupation of the respondents, thereby indicating that the occupation of respondents had no effect/influence on majority of the determinants of their attitude towards waste segregation. In contrast, none of the determinants of PBC, SN & IN exhibited significant association with occupation of the respondents, thereby confirming that respondent's behavior, subjective norms & intention towards waste segregation was irrespective of their occupation.

**5.2.3.6. Effect of study region/city.** Significant association of the responses received for each determinant of attitude was detected, thereby confirming that the waste segregation attitude of the respondents of Jaipur city varied significantly from that of Indore city. The difference in opinion of respondents from both cities for each determinant has been summarized in Table 8. Similar results were noted for all the determinants of perceived behavior control, subjective norms and intention towards waste segregation (Table 8).

### 5.3. Structural model testing

In the proposed structural model, relationship between 4 attributes is proposed on the basis of theory of planned behavior and the hypothesized relationship of the 3 independent attributes (AT, PCB, SN) with the dependent attribute (IN) was tested using PLS-SEM bootstrapping. As shown in Fig. 6, the reflective measurement model showed p value less than 0.05 for all the determinants of their respective latent variables, thus indicating that all the determinants made significant contribution to their respective attributes. Further, the path coefficients of each of the independent variable with that of dependent variable were positive, thus indicating positive relationship of AT, PBC and SN with IN. This positive relationship was further validated with the p value, which was less than 0.05 for all the paths of the model as shown in Fig. 7 and Table 9, thus validating the proposed H1, H2 & H3 hypotheses. Further,

**Table 8**  
City-wise most prominent responses for each determinant of latent variables.

| Attributes<br>(Latent<br>variable) | Determinants | Frequency % (Response)                  |                                        |
|------------------------------------|--------------|-----------------------------------------|----------------------------------------|
|                                    |              | Jaipur city                             | Indore city                            |
| Attitude (AT)                      | AT1          | 51.6 (Important)                        | 80.8 (Important)                       |
|                                    | AT2          | 90.6 (Dissatisfied)                     | 91.7 (Satisfied)                       |
|                                    | AT3          | 54.2 (No impact)                        | 96 (Reduction)                         |
|                                    | AT4          | 66.1 (No/less effective)                | 80.8 (Effective)                       |
|                                    | AT5          | 62.7 (No/less effective)                | 79.4 (Effective)                       |
|                                    | AT6          | 54.2 (Poorly)                           | 90.4 (Well)                            |
|                                    | AT7          | 55.1 (Online resources)                 | 38.2 (All interventions)               |
|                                    | AT8          | 90.6 (Disagree)                         | 79.5 (Agree)                           |
|                                    | AT9          | 62.7 (Negative)                         | 90.4 (Positive)                        |
| Perceived Behavior Control (PBC)   | PBC1         | 55 (Lack of motivation)                 | 43.3 (Inadequate facilities)           |
|                                    | PBC2         | 57.1 (All of the above)                 | 43.7 (All of the above)                |
|                                    | PBC3         | 54.2 (Inadequate training for staff)    | 35.3 (Lack of policy enrollment)       |
|                                    | PBC4         | 64 (Lack of support)                    | 37 (Lack of support)                   |
|                                    | PBC5         | 90.6 (Poor)                             | 50.4 (Good)                            |
|                                    | PBC6         | 64 (Lack of space)                      | 37 (Lack of space)                     |
|                                    | PBC7         | 54.2 (No preference)                    | 46.2 (Dual-stream recycling)           |
| Subjective Norms (SN)              | SN1          | 54.9 (Never)                            | 43.3 (Occasionally)                    |
|                                    | SN2          | 57.8 (Never)                            | 46.2 (Weekly)                          |
|                                    | SN3          | 55.1 (Not sure)                         | 79.5 (Yes)                             |
|                                    | SN4          | 51.6 (Providing resources & facilities) | 41.1 (Education & awareness campaign)  |
|                                    | SN5          | 54.5 (Social media)                     | 44 (Social media)                      |
|                                    | SN6          | 54.2 (Never)                            | 50.4 (Occasionally)                    |
|                                    | SN7          | 64.1 (Slightly active)                  | 43.1 (Moderately active)               |
| Intention (IN)                     | IN1          | 54 (Financial incentives)               | 36.8 (Improved segregation facilities) |
|                                    | IN2          | 62.5 (Disagree)                         | 80.8 (Agree)                           |
|                                    | IN3          | 57.6 (Local competitions)               | 41.1 (Regular feedback)                |
|                                    | IN4          | 54.2 (Legal requirements)               | 36.8 (Social pressure)                 |
|                                    | IN5          | 45.3 (Community based incentives)       | 38.2 (All options)                     |
|                                    | IN6          | 51.3 (Monetary rewards)                 | 46.2 (Recognition/awards)              |
|                                    | IN7          | 48.4 (Regular solid waste)              | 53.1 (Regular solid waste)             |

overall fitness of the model was determined on the basis of  $R^2$  and adjusted  $R^2$  values which were 0.860 and 0.859, respectively.  $R^2$  value higher than 0.5 and closer to 1.00 is recommended for a model to be considered as good fit (Razali et al., 2020), and therefore the model proposed in this study was a good fit and indicated that the independent variables (attitude, perceived behavior control, subjective norms) explained the dependent variable (intention) by ~86 %, thus indicating a good fit model (Table 9).

#### 5.4. Comparison of structural model of Jaipur and Indore

As illustrated in previous sections, that all the determinants of independent as well and dependent variables showed significant association with the location/city of residence. Therefore, the proposed structural model was also tested separately for both Jaipur and Indore cities in order to determine the variations in the overall effect of the independent variables on dependent variables.

Several determinants of the variables were found to have p value higher than 0.05, which indicate that such determinants did not contribute to their respective attributes/latent variable significantly.

From 9 determinants of AT variable, 2 (AT5,  $p = 0.437$  & AT9,  $p = 0.437$ ) of them had p values higher than 0.05 in measurement model of Jaipur city, while in that of Indore city, 4 determinants (AT2,  $p = 0.976$ ; AT6,  $p = 0.976$ ; AT8,  $p = 0.589$  & AT9,  $p = 0.976$ ) had p values higher than 0.05. From 7 determinants of PBC variable, PBC1 ( $p = 0.479$ ) and PBC2 ( $p = 0.984$ ) from Jaipur city model and PBC1 ( $p = 0.060$ ), PBC2 ( $p = 0.393$ ), PBC3 ( $p = 0.062$ ), PBC5 ( $p = 0.834$ ) and PBC7 ( $p = 0.728$ ) from Indore city model had  $p > 0.05$ . Further from 7 determinants of SN and IN, all determinants of SN (except SN4 & SN6) & IN (except IN4, IN6 & IN7) from Jaipur city model and 3 determinants of SN (SN2, AN4 & SN6) and IN (IN3, IN5 & IN6) from Indore city model have  $p > 0.05$ , thus indicating that these determinants did not have any significant contribution to their respective latent variables (Fig. 8ab and Table 10).

Based on the path coefficients, positive relation of AT (0.850) and SN (0.285) and negative relation of PBC (-0.146) with IN in Jaipur city model, whereas in Indore model, positive relation of AT (1.339) and PBC (0.852) and negative relation of SN (-1.078) with IN was noted (Fig. 7ab). In Jaipur city structural model, significant relation of IN in terms of p value was established only with AT and SN, while in case of Indore structural model, significant relation of IN (in terms of p value) was established with AT and PBC, thus indicating that out of the 3 independent variables only 2 contributed significantly and had positive effect on intention in both cities (Fig. 7ab). Further, the adjusted  $R^2$  value of Jaipur city and Indore city was 0.948 and 0.987, respectively which was very close to absolute value (1.00), thus indicating a very good fit of models obtained for both the cities. However, AT & PBC of Indore city model explained the intention by ~98.7 %, while AT & SN of Jaipur city model explained intention by slightly lesser i.e. 94.8 % (Fig. 8ab and Table 10).

## 6. Discussion

This study aimed to identify the effect of 3 key parameters of TPB (attitude, perceived behavior control and subjective norms) on intention of residents of Jaipur and Indore cities towards waste segregation at household level. Further, tests were run to test the proposed hypotheses and model to establish relationship of the 3 key parameters of TPB, with intention.

The socio-demographic profile of the sample population selected in the present study did not exhibited any biasness and for each of the parameter the number of respondents was proportionate which ruled out the possibility of biasness in responses to the questionnaire construct due to their age, gender, marital status and education level. Studies have indicated that, male and female respondents hold divergent perspectives on situational elements that can either motivate or hinder their engagement in solid waste separation practices at home, wherein the education & employment variables exerted societal pressure on individuals to segregate their food waste (Ghani et al., 2013). However, in the present study the sample population had homogenous socio-demographic characteristics, thus eliminating any such cases of opinion biasness due to presence of high proportion of respondents having same gender, age, education and occupation status. It is also mention worthy that the demographic characteristics of sample population from both cities were not statistically different which further ensured the homogeneity among the sample population of Jaipur and Indore cities.

The reliability and validity of the survey instrument showed that all the determinants used described their respective variables very well and were thus reliable and were found to have good internal consistency. The scores of CA, CR, AVE for attitude, perceived behavior control, subjective norms and intention & outer loadings for their respective determinants were clearly within the acceptable/standard benchmark limit of 0.7, 0.7, 0.5 and 0.6, respectively. These scores were in line with those presented in similar previous studies by Vijayan et al. (2023), Razali et al. (2020) and Shi et al. (2021).

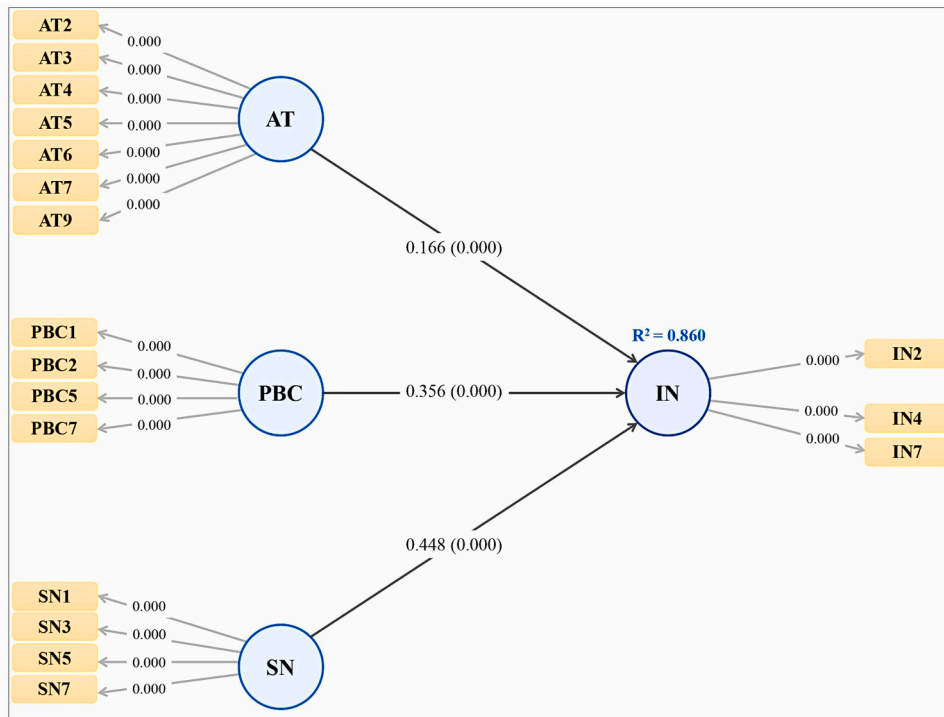


Fig. 7. PLS-SEM showing structural and measurement model statistics.

**Table 9**  
Structural model statistics.

| Hypothesis                          | PC    | t-value | p-value | R <sup>2</sup> | Hypothesis supported |
|-------------------------------------|-------|---------|---------|----------------|----------------------|
| H1: AT is positively related to IN  | 0.166 | 4.035   | 0.000   | 0.860          | Supported            |
| H2: PCB is positively related to IN | 0.356 | 10.367  | 0.000   |                | Supported            |
| H3: SN is positively related to IN  | 0.448 | 17.826  | 0.000   |                | Supported            |

Further, effect of socio-demographic parameters on the attitude, perceived behavior control, subjective norms and intention of the respondents towards waste segregation in their home was also assessed and no significant effect of any of the parameters including age, gender, marital status, education and occupation was found. However, all the 4 attributes varied significantly among respondents belonging to different cities, in other words no other socio-demographic parameter except the respondent's city of residence had significant effect on their attitude, perceived behavior, subjective norm and intention towards waste segregation. The variations in all these 4 attributes of the respondents from different cities was also evident in their response pattern for each questionnaire item. Thus, it could be said that the difference in opinion of respondents was only due to their city of residence and was independent of other socio-demographic variables. In case of attitude, Indore city respondents showed a comparatively more positive response than respondents from Jaipur city in terms of (i) their satisfaction for waste collection/segregation services, (ii) effectiveness of education programs/materials on waste segregation, (iii) awareness about the pros of waste segregation at home, and (iv) implications of waste segregation on health & environment. In case of perceived behavior control, majority of the Jaipur city respondents stated lack of motivation as main reason for practicing waste segregation at home while those of Indore city stated availability of inadequate facilities as the reason. Similarly contradicting opinions were also recorded for the major challenges in implementing waste segregation at academic organizations. In fact, the waste

segregation facility was rated poor by residents of Jaipur city as opposed to "good" rating by residents of Indore city. Further, lack of information about different waste segregation methods among respondents of Jaipur city in comparison to those of Indore city was also noted. In case of subjective norms, drastic difference of opinion in promotion of waste segregation practices or receiving information about waste segregation from any source, role of local government and non-government organizations in implementing waste segregation was noted. The opinion differences clearly indicated that in Jaipur city no significant methods/measures were adopted by the stakeholders for effective waste management, wherein case of Indore city several measures at various levels were being implemented on a regular basis. In case of intention, respondents from Jaipur emphasized more on incentivization in terms of monetary rewards or financial support for promoting/increasing waste segregation at household level, while those from Indore believed in improving the segregation facilities by taking regular feedbacks and using rewards/recognition for promoting waste segregation. The difference in AT, PBC, SN and IN of residents from Jaipur and Indore cities thus explains the drastic difference in the Swachhta rankings of the two cities.

Studies have indicated that TPB offers a valuable framework for examining the elements that impact the choices made by homeowners in waste separation (Sahoo et al., 2022; Shi et al., 2021; Vijayan et al., 2023). As per the model proposed in present study, positive relation of all 3 TPB components was established with intention towards waste segregation. Intention for waste segregation at home was found to be significantly affected with the attitude, perceived behaviour control and subjective norms towards waste segregation, thus all 3 components emerged as prominent predictors of intention for waste segregation in the present study.

According to Pakpour et al. (2014), attitude (AT) refers to the extent to which performance is either positively or negatively evaluated and in the present study it indicated the tendency of residents to engage in waste sorting behaviours. Intention (IN) have been rendered as personal inclination to engage in a particular behaviour, typically seen as a direct factor and prerequisite for actual behavior (Cristea and Gheorghiu, 2016). Ajzen (1991) asserted that AT is a highly influential predictor of

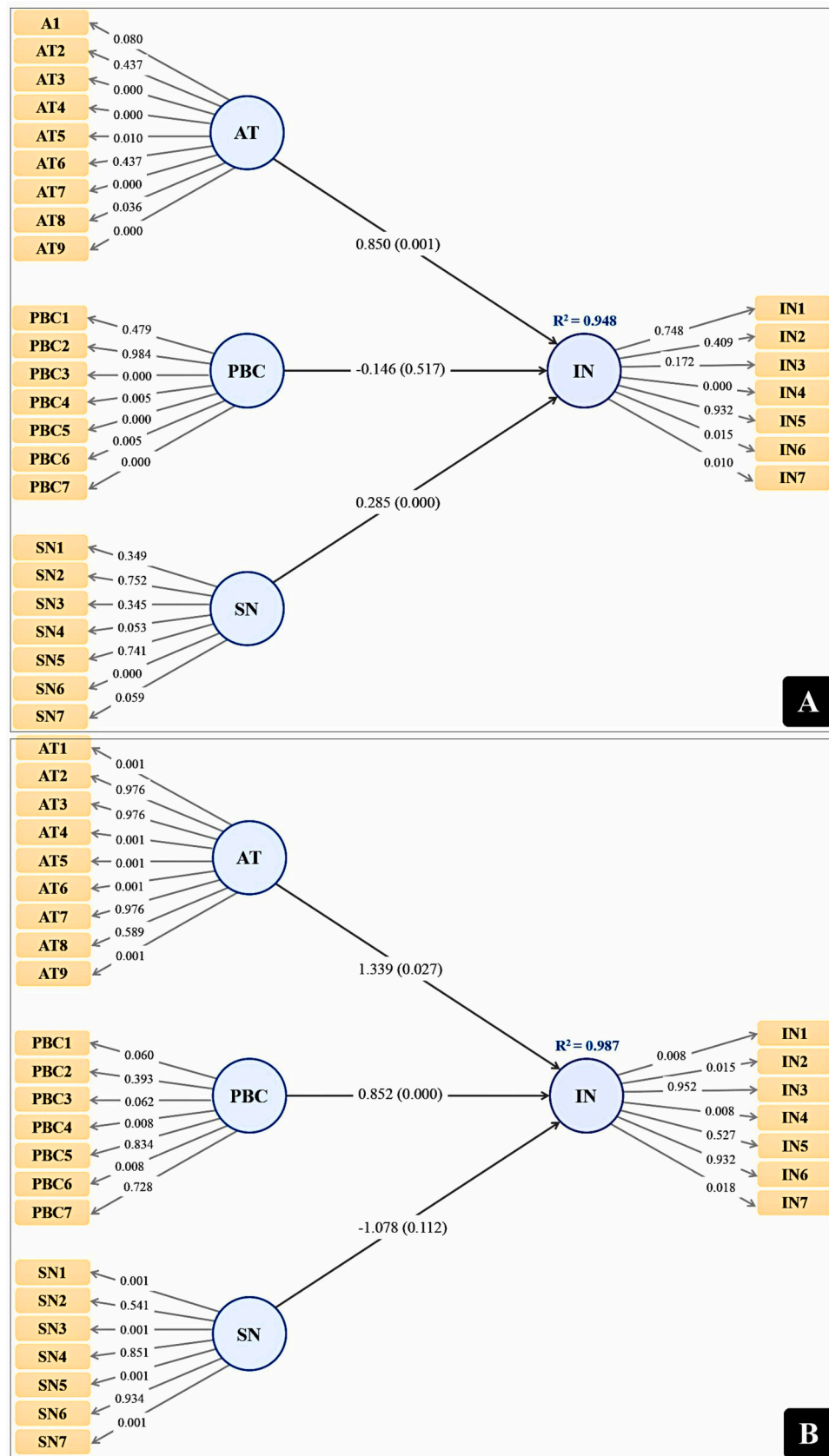


Fig. 8. PLS-SEM model for (A) Jaipur and (B) Indore showing structural and measurement model statistics.

**Table 10**  
Comparison of structural model statistics of Jaipur city and Indore city.

| City                                       | PC     | t-value | p-value | R <sup>2</sup> | Hypothesis supported |
|--------------------------------------------|--------|---------|---------|----------------|----------------------|
| <b>H1: AT is positively related to IN</b>  |        |         |         |                |                      |
| Jaipur                                     | 0.850  | 3.425   | 0.001   | 0.948          | Supported            |
| Indore                                     | 1.339  | 2.217   | 0.027   | 0.987          | Supported            |
| <b>H2: PCB is positively related to IN</b> |        |         |         |                |                      |
| Jaipur                                     | −0.146 | 0.647   | 0.517   | 0.948          | Not supported        |
| Indore                                     | 0.852  | 3.844   | 0.002   | 0.987          | Supported            |
| <b>H3: SN is positively related to IN</b>  |        |         |         |                |                      |
| Jaipur                                     | 0.285  | 3.798   | 0.000   | 0.948          | Supported            |
| Indore                                     | −1.078 | 1.591   | 0.112   | 0.987          | Not supported        |

IN. In fact, many studies have indicated that attitude towards environment significantly impacts the intention (Gkargkavouzi et al., 2019; Shi et al., 2019) and similar relation between attitude and intention *w.r.t.* waste segregation was obtained in the present study. Other than AT, PBC which represents an individual's subjective assessment of their capability to engage in a particular behavior (Ru et al., 2018) and SN which represents perceived external pressure from people around or society (Jabeen et al., 2019) are also known as core predictors of intention (IN). In a similar previous study, authors reported that "sorting waste behaviours were decided the most by individual willingness, which was evident from relatively high magnitude of AT→IN path" (Shi et al., 2021). Contrastingly, Shen et al. (2019) in their study reported that there was no noteworthy link between BA (behavioural attitude) and BI (behavioural intention) while investigating the main factors that influence the municipal waste sorting intents and behaviours of young individuals of Hebei Province in China. However, in the present study, the path magnitude (in terms of coefficient) was comparatively lowest for path AT → IN, thereby indicating least contribution of AT to the waste sorting intentions. It is noteworthy that though AT had least contribution to the waste sorting intentions, yet it had a significant positive effect. Magnitude of paths PBC → IN and SN → IN were relatively higher in the present study and infact that of later was highest, thus indicating that SN which represents the level of societal influence on participants to practice waste separation in their households contributed most to the waste segregating intentions of the respondents. This could be attributed to the fact that the respondents were influenced by someone in their immediate circle rather than making their own independent decision. SN have been rendered as a vital predictor in pro-environmental behaviours including selecting green products (Verma and Chandra, 2018; Yadav and Pathak, 2016), sustainable behaviours of private sectors (Zheng et al., 2018), green retrofitting of buildings (He et al., 2019), pollution reduction preferences (Cordano and Frieze, 2000) and all of these studies have reported direct positive effects SN and BI. Similar to present study, Gkargkavouzi et al. (2019) also reported SN as the most significant determinant of intention with a relatively high path coefficient. Similar to findings of present study regarding significant positive effect of PBC on IN, Shi et al. (2021) and Pakpour et al. (2014) reported that "PBC played key role in decision making regarding sorting waste intention and behaviour" and thus it predict both attitude and intention significantly. Several other studies related to assessment of pro-environmental behaviours also reported similar findings (Holdsworth et al., 2019; Ru et al., 2018; Sun et al., 2019; Zheng et al., 2018). Liu et al. (2018) in their study on consumer's green behaviour using extended TPB, highlighted that consumers are more likely to have a strong intention if they believe they have significant influence over their behaviours. Ru et al. (2018) in their report suggested that citizens will be more inclined to participate in certain behaviours if they have a high level of confidence in their ability to regulate and execute those behaviours. Therefore, based on these facts, it can be said that in case of waste sorting, residents are more likely to engage in sorting behaviours if they find it easy to classify and have access to recycling containers. Furthermore, the theory of organisational behaviour suggested that intention and behavioural control are the primary factors that can

predict final behavior (Hersey et al., 2013). From an individual standpoint, the translation of behavioural intention into actual behaviour may not be fully realized due to the limitations or postponement caused by perceived behavioural control (PBC). Individuals with exceptional aptitudes or well-developed expertise exhibit a pronounced inclination to engage in the specified actions.

In the present study, the proposed model was tested separately for both Jaipur and Indore cities and effect of the TPB attributes were compared. For both cities, only two of the three attributes showed significant positive effect on waste sorting intentions such that for Jaipur city those attributes were AT and SN, while for Indore city they were AT and PBC. In both cases attitude had most prominent/decisive effect on waste segregation intentions of households, while the other attribute had comparatively less effect. Maximum contribution of attitude in both models confirmed that in both cities the waste sorting intentions of residents was mainly influenced by their willingness, wherein other factors involved societal influence in case of Jaipur city and perceived control in case of Indore city. Overall, findings of the present study were in line with the assumption of Ajzen (1991), which stated that "people's given behaviors depend on deep considerations and their readiness".

## 7. Conclusion and future implications

Present study focused on determining the factors influencing waste sorting intentions of residents from two different cities, Jaipur city (ranked 1st in cleanliness across state) and Indore city (ranked 1st in cleanliness across country) through theory of planned behaviour. Findings showed no significant influence of any of the socio-demographic parameters like gender, age, education and occupation on any of the 4 waste sorting attributes (AT, PBC, SN & IN), however significant variations in attitude, perceived behaviour, subjective norms and intentions was obtained among respondents from different cities. The variations further illustrated that in Jaipur city only attitude and subjective norms had positive significant effect on waste sorting intention, while in case of Indore city only attitude and perceived behaviour had significant positive effect on waste sorting intentions. The variations in opinions of respondents from different cities clearly highlighted the need for changes/improvement in strategies for spreading awareness about importance of waste segregation in terms of its pros on health and environment, effective implementation of waste segregation not only at households but also at educational institutes, adoption of different means for promoting waste segregation such as incentivization, rewards/recognitions (e.g. Swachhta rankings), availability of more sources, promotion by influential persons and many more at local community levels. The limited geographical scope of this study may restrict the applicability of the results to other areas. Overall, the proposed model revealed significant positive effect of all 3 attributes (AT, PBC & SN) on waste sorting intentions at household with maximum effect of SN followed by that of PBC and AT. Based on these findings, it could be said that since SN was found to have most prominent effect on intention of residents, which implies that intention of people for sorting of waste at their homes depends greatly on the opinions of others and thus strategies could be developed keeping in consideration these points. Overall, present study proved the hypothesis that individuals' intentions (IN) are contingent upon their attitude (AT), subjective norms (SN), and perceived behavioural control (PBC), and they serve as the determining and fundamental components in the proposed model/theory.

Future research could expand on this study by exploring waste segregation behaviors across a wider range of urban and rural contexts in India, helping to identify regional variations and commonalities in waste management practices. Longitudinal studies would be valuable in understanding how these behaviors and attitudes evolve over time, especially in response to targeted interventions or policy changes. Incorporating additional behavioral theories, such as the Health Belief Model or Social Cognitive Theory, could offer deeper insights into other factors influencing waste segregation practices. Further exploration of

cultural and social norms across different regions would help develop more culturally sensitive waste management strategies. Additionally, investigating the impact of specific policy measures, educational campaigns, and technological advancements, such as smart waste bins or mobile apps, could inform the design of more effective interventions. Further study could investigate the psychological and behavioural mechanisms that contribute to waste segregation intentions, such as the impact of intrinsic motivation and external influences. Also, conducting comparative research on cities with varying waste management techniques and legislation can provide valuable insights into the most effective methods and identify areas that need improvement. This research would ultimately contribute to a more comprehensive understanding of the factors driving sustainable waste management practices in India.

## Abbreviations

AT – attitude, PCB- perceived behavioural control, SN – subjective norms, IN - intention, TBP - theory of planned behaviour

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## CRediT authorship contribution statement

**Maheshchandra Babu Jampala:** Investigation, Data curation and Formal analysis, Writing – original draft. **Tina Shivnani:** Writing – review & editing, Methodology, Conceptualization.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.clwas.2024.100188.

## Data availability

Data will be made available on request.

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